Fiscal Year:	FY 2016	**Task Last Updated:**	FY 01/03/2017
PI Name:	Nelson, Gregory A. Ph.D.		
Project Title:	NSCOR: Charged Particle Radiation and Resultant Oxidative Stress Elicit Deleterious Functional Changes in the Central Nervous System		
Division Name:	Human Research		
Program/Discipline:	HUMAN RESEARCH		
Program/Discipline--Element/Subdiscipline:	HUMAN RESEARCH--Radiation health		
Joint Agency Name:		**TechPort:**	No
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Human Research Program Risks:	(1) **BMed**:Risk of Adverse Cognitive or Behavioral Conditions and Psychiatric Disorders		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Comments:			
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Flight Assignment:	NOTE: Extended to 6/30/2016 per D. Risin/JSC, S. Monk/LaRC, and NSSC (Ed., 1/5/16) NOTE: Extended to 12/31/2015 per PI and NSSC information (Ed., 11/14/14)		
Key Personnel Changes/Previous PI:			
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Performance Goal No.:	
Performance Goal Text:	
Task Description:	The proposed project will investigate responses of the mouse brain to several radiation exposure regimes corresponding to galactic cosmic rays and solar particle events. The project will characterize behavioral, electrophysiological, biochemical, and structural changes both in vivo and in vitro. Radiation-elicited oxidative stress mechanisms will be investigated as important mediators of the biological responses in both wild type and mutant animals. In addition to characterization of normal tissue responses, the potential enhancement of three late risks by radiation exposure will be explored. These are seizures, ischemic injury, and traumatic injury. Mathematical models of the neurotoxicity responses will be developed for use in estimating risks and fitted to project-specific data and previously-published observations. The aims of the proposal will address Human Research Program (HRP) risk 1.2 and at least five of the specific gap questions identified by NASA: Central Nervous System (CNS)-1, -3, -5, -6, and -11.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Understanding the responses of the central nervous system to charged particle radiation has two main benefits, risk assessment and optimization of radiotherapy. The general population and aviation crews are exposed to charged particles during cross country and trans-oceanic air travel at high altitudes as well as to radon in their homes and workplaces. While these exposures are not as high as those of astronauts they are not insignificant. Cancer risks have traditionally been the focus of health risk estimates but the new awareness of sensitive targets in the brain such as neurogenic centers in the hippocampus raises question about the negative effects of such radiation on late neurodegenerative diseases and potential decline of cognitive abilities. This project directly addresses cognitive and neurogenesis issues and indirectly addresses late neurodegenerative effects that share reactive oxygen chemistry with radiation as underlying mechanisms (Alzheimer's, Huntington's, and ALS diseases, are of interest). In the last ten years, the use of charged particles in radiation therapy has grown dramatically with proton and carbon ion treatment centers being established in the U.S., Europe, and Asia. Always limiting the effectiveness of radiotherapy is the response of normal tissue near the tumor site. The precise dose localization associated with charged particle therapy makes it ideal for tumors of the head and neck where careful avoidance of critical structures is critical. Knowledge of the dose tolerance of brain tissue and the reactions to relatively low doses is critical in optimizing these modalities of treatment.
	Introduction. The NSCOR project characterized responses of the central nervous system to charged particles along four major lines of investigation: 1) structural changes to tissue and neurons, 2) electrophysiological measures of function, 3) behavioral alterations, and 4) radiation-induced oxidative stress and inflammation-related responses. Highlights from these investigations are presented below. Structural Changes. The functions of the brain are determined by its structural components, their inter-relations and their dynamic regulation in response to experience and the environment. We examined changes in tissue composition and cell structure to provide insight into the responses of the brain to radiation exposure. The hippocampus is a site of continued cell division leading to the generation of new neurons and glia (neurogenesis) throughout life in vertebrates including humans. We extended previous investigations with charged particles and showed that the production of newly-born neurons and glia was inhibited by all particle types tested to doses >= 0.5 Gy accompanied by unique patterns of differentiation including inhibition of oligodendrocyte production not seen with x-rays or gamma rays. Accompanying the impaired neurogenesis was inactivation of microglia suggesting increased oxidative stress, altered innate immunity, or neuroinflammation. The degree of neural precursor (stem) cell growth, replication, and differentiation were found to be correlated to behavioral impairments and expression of the immediate early gene, Arc, which marks behaviorally-stimulated neurons. Responses were ion-specific and neural field specific and exhibited non-linear dose responses. Structural analysis on neurons at the microscopic level was carried out using Tg(Thy1-EGFP)MJrs/J transgenic mice which express eGFP (enhanced green fluorescent protein) in specific subsets of neurons which are brightly fluorescently labeled for morphometric analyses. We first characterized neuron micromorphology after low doses of protons (0, 10, 100 cGy) which elicited significant reductions in dendritic complexity and spine density along with major alterations in synaptic protein levels. These observations were then extended to other charged particle species (16-O, 28-Si, 48-Ti) at lower doses (e.g., 3 cGy), and longer times (6 months), and the reductions in dendritic complexity and spine density were found to be persistent under these conditions. Dendritic spine analysis using Golgi-stained preparations rather that eGFP transgenic animals confirmed that 56-Fe irradiation alters spine density and dendritic complexity in the mouse hippocampus. Importantly, behaviorally-stimulated enhancement of spine density of basal dendrites in the CA1 region of the hippocampus was blocked by Fe irradiation. Together the observed structural changes would be expected to impair neuronal computation and information transfer. The effects of irradiation on spine morphology are likely indicative of cognitive injury to the CNS, as dendritic spines are receiving excitatory input and changes in their morphology are associated with functional synaptic alterations. Consistent with this notion, abnormalities in dendritic spines morphology have been described in several neurological disorders associated with cognitive impairments, including age-related cognitive decline in the absence of frank dementia, Alzheimer's disease, Huntington's disease, schizophrenia, and neurodevelopmental disorders, including autism spectrum disorders. Electrophysiological Responses. The fundamental property of neurons is their ability to receive, integrate, and transmit electrochemical signals in circuits used for movement, sensory perception, and cognition. We used in vitro electrophysiological recordings in isolated mouse and rat brain slices to measure the effects of protons and HZE (high energy) particle radiation on synaptic transmission and neuronal excitability in rodent hippocampus and medial prefrontal cortex. Most recordings were performed at 30 and 90 days post irradiation with protons, silicon, and iron ions at doses of 10 to 100 cGy in hippocampal excitatory CA1 neurons (glutamatergic Schaffer collateral-CA1 synaptic pathway) and in the granular cells (GC) of the dentate gyrus, DG (medio-lateral perforant pathways-to GC). The results from DG granule cells indicated that the impact of different types of radiation on synaptic parameters may differ (pre- versus postsynaptic effects) and yield different outcomes when compared to the CA1 hippocampal field. The results mainly indicated that it is the synaptic connectivity that is affected by radiation. In the DG, proton exposure increased synaptic excitability which was opposite to the response observed in the CA1 neuronal field. These results

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suggest that proton irradiation reduced GABA-ergic synaptic transmission. There was a striking difference between effects of Si and proton radiation in the DG neurons as Si irradiation did not affect synaptic excitability. However, after iron irradiation there was increased glutamatergic synaptic. This highlights the regional and ion-specific nature of the electrophysiological responses.

In CA1 neurons we observed chronic alterations in several intrinsic passive and active neuronal membrane properties which included changes in membrane resting potentials along with reduced input resistance which both tend to decrease the excitability of CA1 pyramidal neurons. Potentially significant impacts on network function were confirmed by a computational model of the CA1 microcircuit. In the same tissue, that proton irradiation upregulated the persistent Na-currents and increased the rate of miniature excitatory post-synaptic currents which contributed to the neuronal depolarization and increased excitatory state of the hippocampal network. These results show that long-term alterations in the intrinsic properties of the CA1 pyramidal cells and possible adaptive responses occur after low dose irradiation with protons. In coordinated studies on behaviorally trained mice we found that cognitive testing may affect the radiation-induced impacts on long term potentiation (LTP), a persistent synaptic plasticity process.

To explore the impact of proton irradiation on the neurotransmission of specific circuits, paired recordings (accompanied by proteomics and super-resolution imaging) were undertaken in the hippocampus. Within the hippocampus, cannabinoid type 1 receptor (CB1)-expressing basket cells (CB1BCs) and parvalbumin (PV)-expressing interneurons (PVINs) provide inhibition to pyramidal cells (PCs). In irradiated cohorts, GABA release from CB1BCs onto CA1 PCs was increased by a factor of six and was abolished by a CB1 blocking agent (AM251), indicating that protons decreased CB1-dependent tonic inhibition of GABA release from CB1BCs. In contrast, while GABA release from PVINs was unchanged, the excitatory connectivity from CA1 PCs to the PVINs, but not to CB1BCs, was significantly increased by irradiation. These results demonstrate that charged particles elicit cell type-specific, long-term plasticity of local inhibitory and excitatory circuits in the hippocampus which would be expected to impair cognitive performance.

Behavioral Responses. A critical measure for the significance of changes to the brain are whether they result in behavioral changes which integrate all aspects of structure and function including direct impairments and compensatory responses. We used a variety of behavioral measures to determine whether low doses of charged particles result in behavioral changes. These focused on memory tasks linked to hippocampus function but also involved inputs and outputs to amygdala and prefrontal cortex. Tests included contextual and cued fear conditioning, novel object and place recognition, and various mazes. Overall we found that low doses of charged particles are capable of eliciting behavioral alterations that evolve with time over several months, are persistent out to at least 6 months post irradiation, show complex dependence on both dose and radiation type but correlate with structural and electrophysiological changes.

Novel object recognition, object in place recognition, and variations of these tests demonstrated that doses of oxygen and titanium ions lower than 10 cGy can cause long term impairments (months) to memory and correlate with dendritic branching and spine density decreases. The ability of exploratory behavior to increase the density of basal dendritic spines in the CA1 region of the hippocampus and the density of spines in the enclosed blade of the dentate gyrus was affected by Fe irradiation. Contextual and cued fear conditioning, which add an element of emotional stress as motivators, also show long term changes and exhibit non-linear dose responses which may exhibit opposing signs (i.e., enhancement vs decrement) between radiation types such as protons and silicon ions. Another example of the complex dependence on radiation type is the observation that there is a profound increase in contextual fear memory a day following post-training after whole body X-ray irradiation.

Observations showed that in behaviorally naïve mice, Si irradiation at 0.25 and 1 Gy enhances the magnitude of long-term potentiation (LTP), a tissue level electrophysiological measure of memory formation. In mice irradiated with 0.25 Gy, contextual fear memory is enhanced and this is associated with a further enhancement of the LTP magnitude. Such increase in synaptic plasticity is not seen in cognitively tested mice irradiated with 1 Gy. Thus, low dose Si irradiation alters synaptic plasticity in the CA1 region of the hippocampus and these effects are modulated by cognitive testing.

Based on pre-irradiation fear conditioning data we designed new studies to assess effects of irradiation on the brain after task learning has occurred. Impairment in extinction of contextual and cued fear following post-training whole body irradiation was detected and post-training gamma irradiation-enhanced contextual fear memory was found to be associated with reduced neuronal activation of the infralimbic cortex as confirmed by a reduction of c-Fos positive cells in the region. (c-Fos expression marks behaviorally-stimulated cells.) Future studies are warranted to determine whether similar effects are seen following space irradiation and whether they are dependent on the interval between radiation exposure and memory testing.

So, a variety of behavioral responses can be persistently modified by charged particle radiation exposure with complex dependencies on dose, radiation type, and time of exposure relative to behavioral experiences. These differences may reflect different sensitivities of brain regions or constituent cells and circuits and highlight the need to use multiple behavioral endpoints in any risk estimation activity. Importantly, behavioral changes correlate with structural and electrophysiological alterations.

Oxidative Stress and Neuroinflammation. Extensive in vitro experiments detailed the redox response of mouse and human neural stem cells (hNSCs) grown in culture as monolayers or as neurospheres. Cultures of hNSCs culture were subjected to 16-O, 28-Si, 48-Ti, and 56-Fe particles using low doses of 0, 5, 25, and 100 cGy and analyzed for oxidative endpoints (reactive oxygen and nitrogen species, antioxidant enzyme expression and mitochondrial function) at days 3, 5, and 7 and weeks 4, 6, and 8 after exposure. Time courses and dose responses were characterized and demonstrated significant, persistent elevation of radiation-induced oxidative stress.

A cross-cutting experimental approach was to compare results in wild type (WT) animals to transgenic MCATtg animals that overexpress a human catalase targeted to mitochondrial. The transgenic phenotype was expected to show reduced oxidative stress due to enhanced catabolism of hydrogen peroxide. This comparison was performed across all endpoints.

A battery of behavioral tests comparing irradiated WT and MCAT animals revealed enhanced hippocampus-dependent memory and reduced anxiety in mice over-expressing human catalase. When we tested the mechanisms of proton-induced changes we found that overexpression of MCAT significantly decreased cognitive dysfunction after proton irradiation and was accompanied by a protective effect against impaired electrophysiological functions including long term potentiation and altered ion channel composition (AMPA receptor subunits). For example, when we compared the effects of proton irradiation in behavioral tests at 1 month post irradiation we observed significant deficits in novel

object recognition and object-in-place recognition in wild type mice which were completely ameliorated in MCAT mice. These data provide compelling evidence that reducing oxidative stress in the irradiated brain has marked neurocognitive benefits. MCAT mice also maintained longer and more complex branching of apical and basal dendritic trees of pyramidal neurons post charged particle irradiation.

Several experiments examined neuroinflammatory processes in the brain post irradiation including single ion and combined irradiation effects on cytokine/chemokine profiles in hippocampal samples one and three months following the exposures. To detect the presence of an adaptive effect animals were irradiated with 10 cGy 150 MeV protons followed 24 hours later by 50 cGy 600 MeV/iron ions. Sham and single ion exposures served as controls. Measurements showed that prior irradiation with protons dramatically modified the subsequent iron-induced changes in behavior and inflammation-related cytokines and chemokines.

Mice irradiated with either protons or protons followed by Fe show an impaired novel object recognition at 3 months, but not 1 month, after irradiation that is not seen in mice irradiated only with Fe. In mice irradiated with protons or Fe alone there is a negative correlation between novel object recognition and the density of newly born activated microglia in the dentate gyrus. Expression levels of several important chemokines/cytokines showed strong modulation by the sequential irradiation versus single ion exposures; these include IL-4, IL-6, IL-12, IFN-g, TNF-a. Particularly striking were effects at three months on macrophage-derived chemokine (MDC) and eotaxin levels which were much lower in mice receiving both types of radiation exposures. The levels of MDC and eotaxin correlate and the levels of MDC further correlate with the percentage of newly born activated microglia in the dentate gyrus.

Because the cytokine profiles indicated a complex response where the small proton priming dose could either potentiate or suppress effects elicited by Fe ions, we examined these responses in more detail using RNA sequencing methods to understand the mRNA as well as the protein expression responses at 7 and 30 days post irradiation. Particular attention was paid to the genes associated with microglia function and differentiation. Increasing evidence indicates that microglial activation in the central nervous system is heterogeneous and can be categorized into two opposite types: M1 phenotype and M2 phenotype. Generally, M1 microglia predominate at injury sites and at the end stage of diseases such as Alzheimers when the resolution of the inflammatory response and repair processes of M2 microglia (alternate pathway) are suppressed. We found several M1-associated genes which showed ion-specific and dual irradiation expression differences. These included: MARCO, NOS-2, and TNF-a. M2-associated genes which showed ion-specific and dual irradiation expression differences included: IFN-g, IL1-b, IL-10, IL-13, CCL2, CCL7, CCL8, CD163, CD206, TGF-b and YM-1. These data show the diversity in gene expression patterns reflecting a complex combination of track structures and particle fluences / doses. They result in simultaneous expression of both microglial phenotypes suggesting a dynamic balance between inflammation and its resolution. Understanding the “rules” underlying these responses will be critical in predicting the outcomes of exposure to the galactic cosmic ray spectrum comprised of many ions and energy.

Conclusions

The NSCOR team has demonstrated that low doses of charged particles (<=10 cGy) can elicit a number of persistent changes to the brain (especially hippocampus). At the structural level, these include reductions in neurogenesis correlated with microglial activation and substantial reductions in dendritic branching and spine numbers. Electrophysiological changes affect both excitatory and inhibitory neurons and include altered intrinsic membrane properties and dysregulation of excitability and synaptic plasticity. Multiple behavioral tests that measure aspects of memory and cognition show impairments in performance. Elevation of oxidative stress is seen soon after irradiation, is persistent, and is accompanied by elevated markers of inflammatory processes. The mechanistic importance of the oxidative stress is shown by the protective effects on all endpoints of elevated catalase activity in MCAT transgenic animals. Together these observations point to a broad-based undesirable reaction of brain tissue to irradiation at doses which do not cause gross pathological changes or cell death. The mechanisms are poorly understood at this time but may involve oxidative mechanisms and inflammatory process. The significance of the impairments in the rodent models to human performance are presently unknown but demonstrate that there is a basis for concern.

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